Rithul Kamesh

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SUMMARY

AI infrastructure and systems engineer building the runtime, storage, and tooling layer beneath LLM and ML workloads. Shipped a multi-tier LLM caching runtime achieving 92.5% token reduction on agent workloads, and an embedded vector database distributed across five language ecosystems, both running in production AI pipelines. Systems background in Rust, Go, C++, and CUDA spanning execution engines, distributed storage, and developer tooling.

SELECTED PROJECTS

Continuum *C++, Python*
Unified LLM and ML Execution Runtime

- Designed a unified IR dispatching LLM token generation and tensor computation through a single C++ execution graph, with a five-tier reuse hierarchy — exact memoization, semantic caching, trie prefix KV caching, layer KV warm-start, and memory graph recall — achieving 92.5% token reduction on mixed agent workloads and up to 99% on shared-prefix calls
- Built provider-agnostic backend support (OpenAI, Azure OpenAI, Anthropic, vLLM, libtorch, Apple MLX) with durable, checkpointable execution enabling process-resumable and forkable agent workflows
- Shipped as `continuum-ai` on PyPI with Python/C++ bindings, CI matrix on Linux and macOS, fuzz workflow, and coverage gates; featured on Product Hunt at launch

Veclite *Rust, Python (PyO3), Go (CGO), C/C++, Zig*
Embedded Vector Database

- Engineered HNSW indexing with SIMD-accelerated distance kernels (cosine, L2, dot product, Manhattan) in Rust with single-file storage and a vector-native SQL dialect supporting hybrid metadata + similarity search
- Maintained 14 stable releases (latest v1.1.1) with an ABI-stable C layer, distributed across 5 language ecosystems — Crates.io, PyPI (GIL-released search via PyO3), Go modules, and C/C++ headers — enabling zero-server, in-process vector search for production AI memory pipelines

Docproc *Python*
Document Intelligence Engine

- Built a document-to-Markdown conversion engine preserving layout, tables, and semantic structure for LLM ingestion pipelines, with vision-model-based figure and diagram description
- Shipped an evaluation harness with a benchmarking suite measuring downstream reasoning fidelity across document types

EXPERIENCE

H2LooP *On-Site*
AI Developer Advocate Intern *July 2026 – October 2026*

- Built technical demos and reference implementations on Hydron's AI-powered embedded development platform, covering firmware generation, debugging, cross-target porting, and hardware-aware code generation workflows
- Produced developer-facing technical content and sample repositories, and ran hands-on workshops introducing AI-assisted embedded workflows to university and professional engineering audiences

TECHNICAL SKILLS

Languages: Rust, Go, Python, C/C++, CUDA
Systems: Distributed Systems, SIMD, Lock-free Concurrency, WGPU, OpenMP, FDTD
Infrastructure: Docker, Kubernetes, Prometheus, Grafana, OpenTelemetry, CI/CD
Backend: REST, gRPC, FastAPI, Redis, Redis Streams, PostgreSQL, pgvector, BM25, WebSockets
AI / ML: PyTorch, Transformers, LangChain, LangGraph, Agent Systems, RAG, LLM Evaluation, HNSW, Vector Databases

EDUCATION

PES University *Bengaluru, India*
B.Tech, Electronics and Communication Engineering *Aug 2023 – May 2027*

BGS National Public School *Bengaluru, India*
HSSC, SSC *Feb 2023*

RESEARCH EXPERIENCE

Photonics and Quantum Technology Lab, PES University *Bengaluru, India*
Undergraduate Research Assistant *Feb 2025 – Present*

- Modeling photonic integrated circuits via Lindblad/Hamiltonian open-quantum-system formalisms and building FDTD-coupled quantum solvers (Crank–Nicholson stepping, CPML boundaries, NTFF transforms) for photonic device characterization and state fidelity analysis

Teaching Assistant, Non-Linear Optics and Quantum Technology *Jan 2026 – May 2026*
Co-delivered lectures and built study guides and question banks for 100+ students.

PUBLICATIONS

B. S. Ullal, V. K. Narasimhaiah, and **R. Kamesh**, "ResNet based deep learning approach for chronic obstructive pulmonary disease prediction using lung sound analysis," *IAES International Journal of Artificial Intelligence (IJ-AI)* (Scopus), vol. 15, no. 2, pp. 1733–1745, Apr. 2026. [ResNet: 97.90% accuracy, 98.72% precision, 0.9937 AUC] doi:10.11591/ijai.v15.i2.pp1733-1745

ACHIEVEMENTS

Hack Night 7.0, Third Place Oct 2025
Speaker, Snowflake User Meetup Chennai Jun 2025
Hack Night 6.0, Second Place Oct 2024